Mahammad Azeez Mogal
Process Improvement Engineer
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Summary

Highly versatile Process Improvement Engineer with 5 years of experience driving operational excellence across manufacturing, digital finance, and customer care domains. Skilled in Lean Six Sigma, Kaizen, and DMAIC methodologies, with a proven track record of reducing process cycle times and improving first-pass yield. Adept at leveraging automation tools (UiPath, Power Automate, Python) and data visualization platforms (Power BI, Tableau) to deliver actionable insights and cost savings exceeding $110K annually. Proficient in cloud platforms (Azure, AWS, ServiceNow), BPM tools, and ITSM frameworks, ensuring regulatory compliance (SOC2, SOX, HIPAA) while enhancing workflow scalability. Recognized for strong analytical thinking, stakeholder communication, and cross-functional leadership to align transformation initiatives with business objectives.

Experience

Process Improvement Engineer | Cognizant, USA – Remote ***Aug 2024 – Present***

- Spearheaded cross-functional Lean Six Sigma initiatives across digital finance and customer care operations, achieving a 20% reduction in process cycle time and a 15% increase in throughput by eliminating over 30 non-value-added steps.
- Integrated Power BI and Tableau dashboards with AI/ML-driven process intelligence tools, enabling real-time KPI monitoring and reducing incident resolution time by 25% across mission-critical workflows.
- Led process automation efforts using RPA tools (UiPath, Power Automate), saving $110K+ annually and improving task accuracy by over 40%, while ensuring alignment with internal compliance and SOC2 audit standards.
- Conducted Agile-aligned process reengineering using BPMN and Value Stream Mapping in collaboration with product owners and tech teams, facilitating seamless integration with cloud-based platforms (ServiceNow, Azure).
- Championed change enablement by designing standardized SOPs, onboarding materials, and training sessions, resulting in a 35% increase in compliance and a 90% adoption rate of reengineered workflows.
- Partnered with cybersecurity and compliance teams to embed controls into redesigned workflows, ensuring risk mitigation and regulatory compliance (SOX, HIPAA) during system transitions.
- Facilitated Kaizen workshops, RCA (Root Cause Analysis), and DMAIC-based continuous improvement sessions, improving first-pass yield from 82% to 93% within six months and boosting cross-team engagement by 50%.
- Influenced executive stakeholders through data storytelling and impact reporting, securing buy-in for process improvement initiatives and aligning transformation roadmaps with strategic business goals.

Mechanical Design Engineer | HCL Technologies Ltd, India ***Jun 2019 – Jul 2023***

- Led the end-to-end mechanical design of semiconductor processing modules including gas boxes, source delivery units, and dual/quad chamber modules with a focus on performance, safety, and manufacturability using Creo and ASME Y14.5 GD&T standards.
- Delivered full 2D/3D drawing packages and RFQ documentation for heater jackets and exhaust ducts, streamlining supplier communication and reducing thermal misalignment issues by 25 percent.
- Applied advanced DFM and DFA principles in the creation of complex sheet-metal and plastic part assemblies, driving an 80 percent reduction in post-prototype rework and minimizing manufacturing iterations.
- Improved engineering change implementation cycles by standardizing drawing templates and automating release workflows in SAP PLM and Teamcenter, cutting design release time by 30 percent.
- Partnered with FEA and CAE teams to optimize mechanical strength and thermal stability, integrating simulation feedback into iterative design loops to enhance product durability.
- Reduced BOM errors and improved traceability by 25 percent through structured configuration control and alignment with ISO 9001 documentation practices.
- Conducted tolerance stack-up studies, interference and clearance checks, and volume optimization for prototype and production builds, ensuring smooth assembly and functional integration.
- Supported sustainability goals by optimizing material selection and part count, reducing raw material usage by 12 percent across multiple designs.
- Mentored junior engineers in 3D modeling, GD&T interpretation, and drawing review practices, improving team productivity and first-pass approval rates.
- Worked on electromechanical assemblies integrating heaters, sensors, valves, cabling, and power routing requirements for vacuum-based semiconductor equipment.
- Designed components for manufacturability involving stamping, machining, and molded-plastic parts, including draft analysis, rib and boss design, and machining tolerances.
- Reviewed stamping feasibility with suppliers, validating bend reliefs, K-factors, and die-clearances for high-accuracy sheet-metal parts.
- Supported molded-part development through resin selection, shrink analysis, gate placement, and thermal-cycling considerations.
- Participated in prototype-to-production transfer aligned with NPI and NPD cycles, preparing 3D models, ECO packages, DFMEA inputs, and supplier documentation.
- Applied tolerance and GD&T decisions to ensure compatibility between stamped housings, machined manifolds, and injection-molded covers.

- Prepared DFMEA risk inputs related to thermal distortion, leak paths, mounting stress, and long-term reliability concerns.
- Selected metals and engineered plastics based on corrosion exposure, chemical compatibility, thermal coefficients, and semiconductor gas-handling requirements.
- Collaborated with teams that operated in Windchill PLM environments while executing document governance in SAP PLM and Teamcenter.
- Supported component qualification and supplier readiness in line with ISO 9001 expectations, drawing approvals, and PPAP-style release checks.
- Presented design intent, tolerance decisions, and risk items to senior engineers, suppliers, and stakeholders during formal design reviews.
- Contributed to service-friendly design choices, fastening strategy, assembly sequence, and ergonomic access for technicians.
- Worked directly with procurement teams during RFQ reviews to evaluate manufacturability, cost drivers, and lead-time risks.

Skills

Process & Quality Improvement: Lean Six Sigma, DMAIC, Kaizen, Root Cause Analysis (RCA), Value Stream Mapping (VSM), SOP Standardization, Continuous Improvement, DFMEA, DFA, ASME Y14.5 GD&T
Process Automation & RPA: UiPath, Power Automate, Ansible (automation scripting), Python (task automation), Netmiko (network config automation)
Data Analysis & Visualization: Power BI, Tableau, Excel (Pivot Tables, Macros), KPI Dashboards, Data Storytelling, Impact Reporting
Workflow Modeling & BPM: BPMN, Microsoft Visio, Process Mapping, Agile Methodologies, Jira, Confluence
Cloud Platforms & ITSM Tools: Azure, ServiceNow, AWS, SOC2, HIPAA, SOX compliance
Monitoring & Observability Tools: ELK Stack, Prometheus, Grafana, SolarWinds, SNMP-based alerting systems
Mechanical Design & CAD Tools: Creo (Pro/E), SolidWorks, AutoCAD, SAP PLM, Teamcenter, exposure to Windchill , ANSYS (FEA), Marlin Firmware
Manufacturing & Design Methods: Stamping, Injection Molding, Machining, Sheet Metal Design, Tolerance Stack-Up, DFM, DFA, DFMEA, Electromechanical Design, NPI/NPD.
Materials: Sheet-metal alloys, machined metals, engineered plastics (ABS, PC, PEEK), thermal and chemical compatibility.
Scripting & Development Tools: Python, Git, PowerShell, Bash
Collaboration & Documentation: Microsoft Office Suite, SharePoint, Teams, Technical Documentation, Training Manuals
Soft Skills: Analytical thinking, cross-functional collaboration, problem-solving, communication, leadership, change management, and time management

Education

Master of Science in Computer Information Systems ***Aug 2023 – May 2025***
Rivier University, Nashua – NH

Bachelor of Technology in Mechanical Engineering ***Aug 2017 – Sep 2020***
Bapatla Engineering College, India

Projects

Fabrication of Multi-Purpose 3D Printer (FDM, CNC, Laser Cutting) | *SolidWorks, ANSYS, Marlin Firmware, DFMEA*
- Led a 4-member team to develop a cost-effective, modular 3D printer with interchangeable toolheads for FDM, CNC, and laser cutting.
- Designed gantry-based mechanical structure in SolidWorks and validated frame stability using FEA in ANSYS, achieving ±0.3 mm precision.
- Configured Marlin firmware for multi-mode operation and integrated a custom SMPS for stable power distribution.
- Applied DFMEA and DFA principles to ensure reliability and ease of assembly; project received top honors at university innovation expo.

Kinetic Energy Recovery System (KERS) Using Rack & Pinion | *AutoCAD, Prototyping, Renewable Energy*
- Designed and built a rack & pinion-based KERS prototype embedded in vehicle suspension to harness braking energy.
- Modeled layout using AutoCAD and integrated a DC generator for energy conversion under varied load conditions.
- Demonstrated potential to generate supplemental power, supporting greener vehicle technology initiatives.

Awards & Leadership

- ERS Champion – Quick Learner (2021), Awarded for mastering tools and early project delivery at HCL
- Live Wire – AHM Award (2022), for consistent quality output and exceeding productivity goals
- Organized and led cultural and technical events like TEDxBEC and BECTAGON2K19 as Cultural Club Coordinator at Bapatla Engineering College

Certifications

- **Six Sigma Green Belt – LinkedIn Learning**